Unit II Data Mining - I

Introduction to Data Mining Systems, Data Mining Techniques – Issues – applications- Data Objects and attribute types, Statistical description of data, Data Preprocessing – Cleaning, Integration, Reduction, Transformation and discretization, Data Visualization, Data similarity and dissimilarity measures.

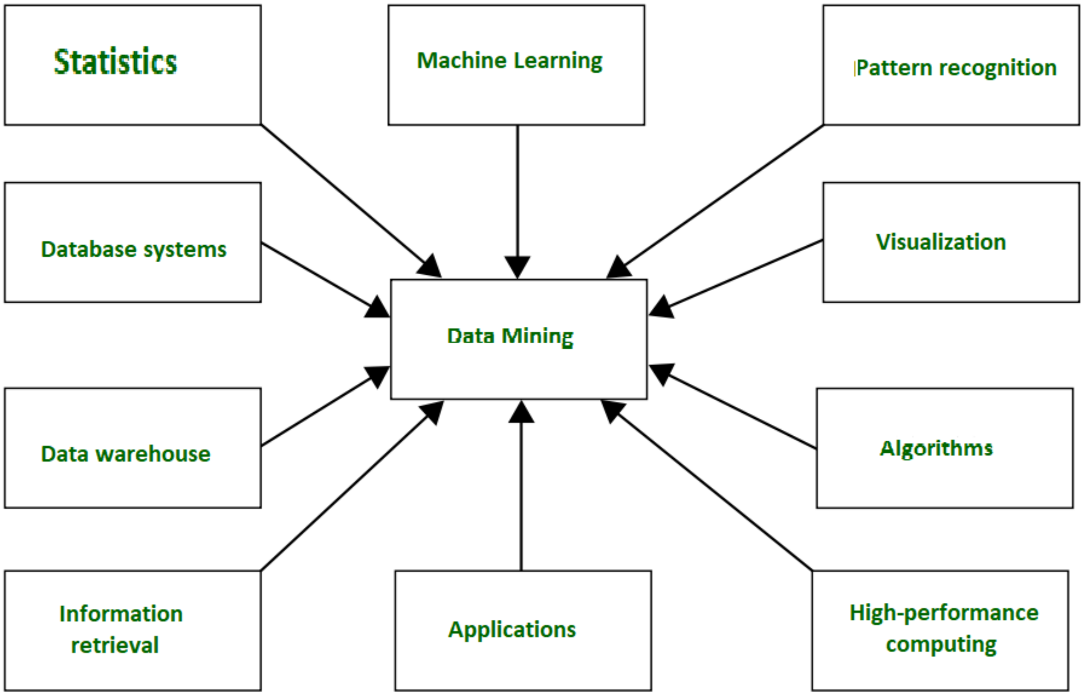
Introduction to Data Mining Systems-

Data mining is the process of extracting useful information from large sets of data. It involves using various techniques from statistics, machine learning, and database systems to identify patterns, relationships, and trends in the data. This information can then be used to make data-driven decisions, solve business problems, and uncover hidden insights. Applications of data mining include customer profiling and segmentation, market basket analysis, anomaly detection, and predictive modeling. Data mining tools and technologies are widely used in various industries, including finance, healthcare, retail, and telecommunications.

In general terms, "**Mining**" is the process of extraction of some valuable material from the earth e.g. coal mining, diamond mining, etc. In the context of computer science, "**Data Mining"** can be referred to as **knowledge mining from data, knowledge extraction, data/pattern analysis, data archaeology, and data dredging**. It is basically the process carried out for the extraction of useful information from a bulk of data or data warehouses. One can see that the term itself is a little confusing. In the case of coal or diamond mining, the result of the extraction process is coal or diamond. But in the case of Data Mining, the result of the extraction process is not data!! Instead, data mining results are the patterns and knowledge that we gain at the end of the extraction process. In that sense, we can think of Data Mining as a step in the process of Knowledge Discovery or Knowledge Extraction.

Nowadays, data mining is used in almost all places where a large amount of data is stored and processed. For example, banks typically use 'data mining' to find out their prospective customers who could be interested in credit cards, personal loans, or insurance as well. Since banks have the transaction details and detailed profiles of their customers, they analyze all this data and try to find out patterns that help them predict that certain customers could be interested in personal loans, etc.

Main Purpose of Data Mining-



Data Mining

Basically, Data mining has been integrated with many other techniques from other domains such as **statistics, machine learning, pattern recognition, database and data warehouse systems, information retrieval, visualization,** etc. to gather more information about the data and to helps predict hidden patterns, future trends, and behaviors and allows businesses to make decisions.

Technically, data mining is the computational process of analyzing data from different perspectives, dimensions, angles and categorizing/summarizing it into meaningful information.

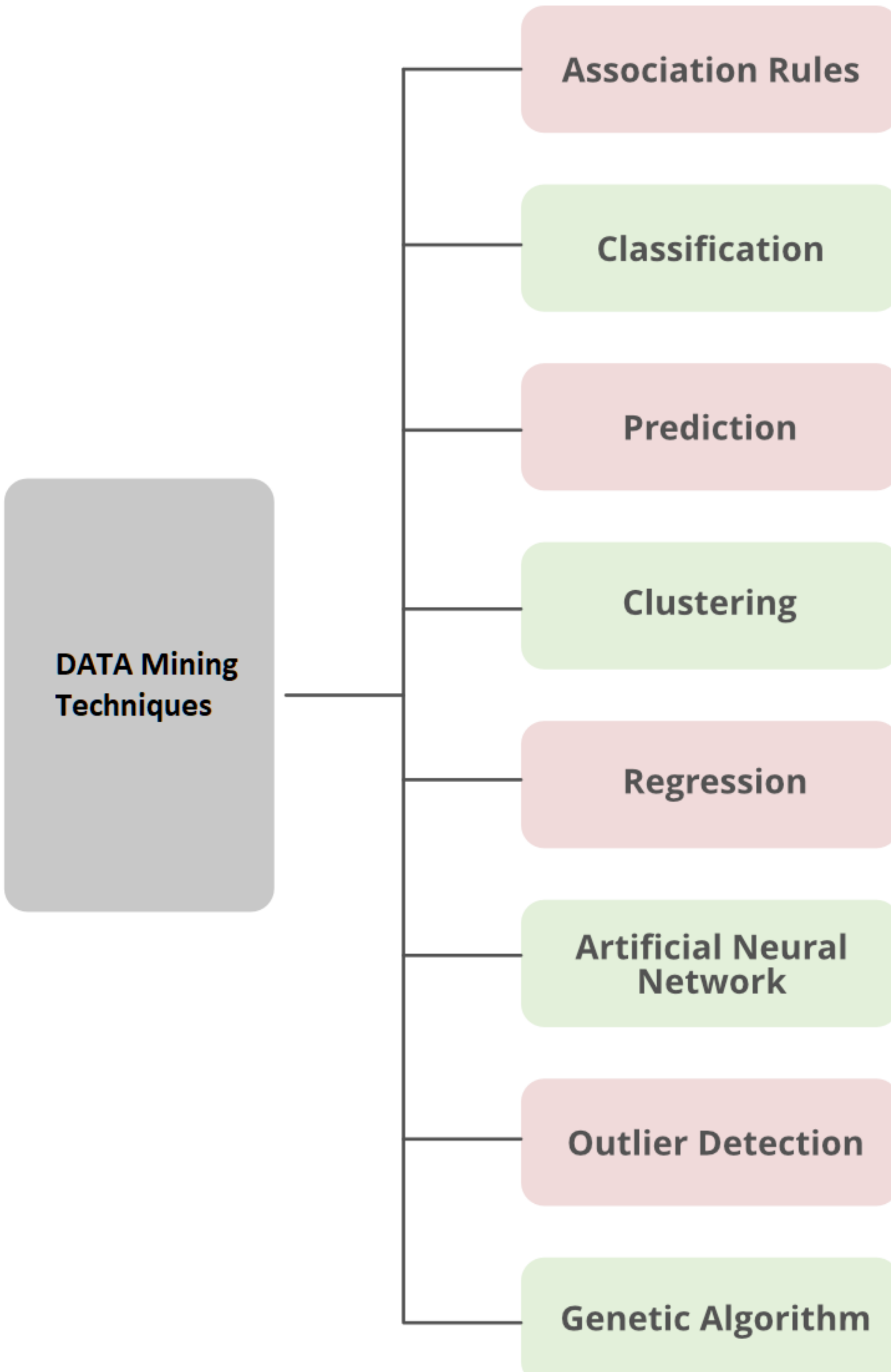
Data Mining can be applied to any type of data e.g. **Data Warehouses, Transactional Databases, Relational Databases, Multimedia Databases, Spatial Databases, Time-series Databases, World Wide Web.**

Data Mining Techniques – Issues – applications-

Techniques-

Data Mining Techniques which are used to predict desire output.

Data Mining Techniques



1. Association

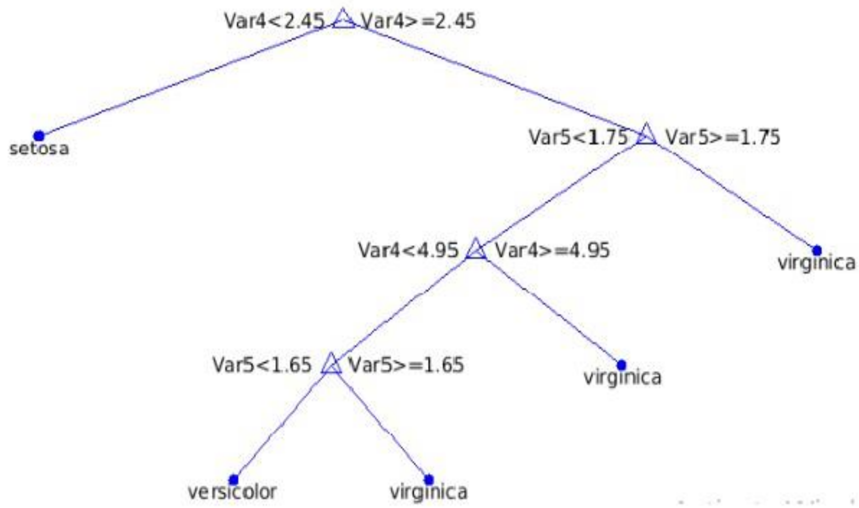
Association analysis is the finding of association rules showing attribute-value conditions that occur frequently together in a given set of data. Association analysis is widely used for a market basket or transaction data analysis. Association rule mining is a significant and exceptionally dynamic area of data mining research. One method of association-based classification, called associative classification, consists of two steps. In the main step, association instructions are generated using a modified version of the standard association rule mining algorithm known as Apriori. The second step constructs a classifier based on the association rules discovered.

2. Classification

Classification is the processing of finding a set of models (or functions) that describe and distinguish data classes or concepts, for the purpose of being able to use the model to predict the class of objects whose class label is unknown. The determined model depends on the investigation of a set of training data information (i.e. data objects whose class label is known). The derived model may be represented in various forms, such as classification (if – then) rules, decision trees, and neural networks. Data Mining has a different type of classifier:

- Decision Tree
- SVM(Support Vector Machine)
- Generalized Linear Models
- Bayesian classification:
- Classification by Backpropagation
- K-NN Classifier
- Rule-Based Classification
- Frequent-Pattern Based Classification
- Rough set theory
- Fuzzy Logic

Decision Trees: A decision tree is a flow-chart-like tree structure, where each node represents a test on an attribute value, each branch denotes an outcome of a test, and tree leaves represent classes or class distributions. Decision trees can be easily transformed into classification rules. Decision tree enlistment is a nonparametric methodology for building classification models. In other words, it does not require any prior assumptions regarding the type of probability distribution satisfied by the class and other attributes. Decision trees, especially smaller size trees, are relatively easy to interpret. The accuracies of the trees are also comparable to two other classification techniques for a much simple data set. These provide an expressive representation for learning discrete-valued functions. However, they do not simplify well to certain types of Boolean problems.



This figure generated on the IRIS (Immune Reconstitution inflammatory Syndrome) data set of the UCI machine repository. Basically, **three different class labels available in the data set: Setosa, Versicolor, and Virginia**.

Support Vector Machine (SVM) Classifier Method: Support Vector Machines is a supervised learning strategy used for classification and additionally used for regression. When the output of the support vector machine is a continuous value, the learning methodology is claimed to perform regression; and once the learning methodology will predict a category label of the input object, it's known as classification. The independent variables could or could not be quantitative. Kernel equations are functions that transform linearly non-separable information in one domain into another domain wherever the instances become linearly divisible. Kernel equations are also linear, quadratic, Gaussian, or anything that achieves this specific purpose. A linear classification technique may be a classifier that uses a linear function of its inputs to base its decision on.

Generalized Linear Models: Generalized Linear Models(GLM) is a statistical technique, for linear modeling. GLM provides extensive coefficient statistics and model statistics, as well as row diagnostics. It also supports confidence bounds.

Bayesian Classification: Bayesian classifier is a statistical classifier. They can predict class membership probabilities, for instance, the probability that a given sample belongs to a particular class. Bayesian classification is created on the Bayes theorem. Studies comparing the classification algorithms have found a simple Bayesian classifier known as the naive Bayesian classifier to be comparable in performance with decision tree and neural network classifiers. Bayesian classifiers have also displayed high accuracy and speed when applied to large databases. Naive Bayesian classifiers adopt that the exact attribute value on a given class is independent of the values of the other attributes.

This assumption is termed class conditional independence. It is made to simplify the calculations involved, and is considered “naive”. Bayesian belief networks are graphical replicas, which unlike naive Bayesian classifiers allow the depiction of dependencies among subsets of attributes. Bayesian belief can also be utilized for classification.

Classification By Backpropagation: A Backpropagation learns by iteratively processing a set of training samples, comparing the network’s estimate for each sample with the actual known class label. For each training sample, weights are modified to minimize the mean squared error between the network’s prediction and the actual class. These changes are made in the “backward” direction, i.e., from the output layer, through each concealed layer down to the first hidden layer (hence the name backpropagation). Although it is not guaranteed, in general, the weights will finally converge, and the knowledge process stops.

K-Nearest Neighbor (K-NN) Classifier Method: The k-nearest neighbor (K-NN) classifier is taken into account as an example-based classifier, which means that the training documents are used for comparison instead of an exact class illustration, like the class profiles utilized by other classifiers. As such, there’s no real training section. once a new document has to be classified, the k most similar documents (neighbors) are found and if a large enough proportion of them are allotted to a precise class, the new document is also appointed to the present class, otherwise not. Additionally, finding the closest neighbors is quickened using traditional classification strategies.

Rule-Based Classification: Rule-Based classification represent the knowledge in the form of If-Then rules. An assessment of a rule evaluated according to the accuracy and coverage of the classifier. If more than one rule is triggered then we need to conflict resolution in rule-based classification. Conflict resolution can be performed on three different parameters: Size ordering, Class-Based ordering, and rule-based ordering. There are some advantages of Rule-based classifier like:

- Rules are easier to understand than a large tree.
- Rules are mutually exclusive and exhaustive.
- Each attribute-value pair along a path forms conjunction: each leaf holds the class prediction.

Frequent-Pattern Based Classification: Frequent pattern discovery (or FP discovery, FP mining, or Frequent itemset mining) is part of data mining. It describes the task of finding the most frequent and relevant patterns in large datasets. The idea was first presented for mining transaction databases. Frequent patterns are defined as subsets (item sets, subsequences, or substructures) that appear in a data set with a frequency no less than a user-specified or auto-determined threshold.

Rough Set Theory: Rough set theory can be used for classification to discover structural relationships within imprecise or noisy data. It applies to discrete-valued features. Continuous-valued attributes must therefore be discrete prior to their use. Rough set theory is based on the establishment of equivalence classes within the given training data. All the data samples forming a similarity class are indiscernible, that is, the samples are equal with respect to the attributes describing the data. Rough sets can also be used for feature reduction (where attributes that do not contribute towards the classification of the given training data can be identified and removed), and relevance analysis (where the contribution or significance of each attribute is assessed with respect to the classification task). The problem of finding the minimal subsets (redacts) of attributes that can describe all the concepts in the given data set is NP-hard.

Fuzzy-Logic: Rule-based systems for classification have the disadvantage that they involve sharp cut-offs for continuous attributes. Fuzzy Logic is valuable for data mining frameworks performing grouping /classification. It provides the benefit of working at a high level of abstraction. In general, the usage of fuzzy logic in rule-based systems involves the following:

- Attribute values are changed to fuzzy values.
- For a given new data set /example, more than one fuzzy rule may apply. Every applicable rule contributes a vote for membership in the categories. Typically, the truth values for each projected category are summed.

3. Prediction

Data Prediction is a two-step process, similar to that of data classification. Although, for prediction, we do not utilize the phrasing of "Class label attribute" because the attribute for which values are being predicted is consistently valued (ordered) instead of categorical (discrete-esteemed and unordered). The attribute can be referred to simply as the predicted attribute. Prediction can be viewed as the construction and use of a model to assess the class of an unlabeled object, or to assess the value or value ranges of an attribute that a given object is likely to have.

4. Clustering

Unlike classification and prediction, which analyze class-labeled data objects or attributes, clustering analyzes data objects without consulting an identified class label. In general, the class labels do not exist in the training data simply because they are not known to begin with. Clustering can be used to generate these labels. The objects are clustered based on the principle of maximizing the intra-class similarity and minimizing the interclass similarity. That is, clusters of objects are created so that objects inside a cluster have high similarity in contrast with each other, but are different objects in other clusters. Each Cluster that is generated can be seen as a class of objects, from which rules can be inferred. Clustering can also facilitate classification formation, that is, the organization of observations into a hierarchy of classes that group similar events together.

5. Regression

Regression can be defined as a statistical modeling method in which previously obtained data is used to predicting a continuous quantity for new observations. This classifier is also known as the Continuous Value Classifier. There are two types of regression models: Linear regression and multiple linear regression models.

6. Artificial Neural network (ANN) Classifier Method

An artificial neural network (ANN) also referred to as simply a "Neural Network" (NN), could be a process model supported by biological neural networks. It consists of an interconnected collection of artificial neurons. A neural network is a set of connected input/output units where each connection has a weight associated with it. During the knowledge phase, the network acquires by adjusting the weights to be able to predict the correct class label of the input samples. Neural network learning is also denoted as connectionist learning due to the connections between units. Neural networks involve long training times and are therefore more appropriate for applications where this is feasible. They require a number of parameters that are typically best determined empirically, such as the network topology or "structure". Neural networks have been criticized for their poor interpretability since it is difficult for humans to take the symbolic meaning behind the learned weights. These features firstly made neural networks less desirable for data mining.

The advantages of neural networks, however, contain their high tolerance to noisy data as well as their ability to classify patterns on which they have not been trained. In addition, several algorithms have newly been developed for the extraction of rules from trained neural networks. These issues contribute to the usefulness of neural networks for classification in data mining.

An artificial neural network is an adjective system that changes its structure-supported information that flows through the artificial network during a learning section. The ANN relies on the principle of learning by example. There are two classical types of neural networks, perceptron and also multilayer perceptron.

7. Outlier Detection

A database may contain data objects that do not comply with the general behavior or model of the data. These data objects are Outliers. The investigation of OUTLIER data is known as OUTLIER MINING. An outlier may be detected using statistical tests which assume a distribution or probability model for the data, or using distance measures where objects having a small fraction of "close" neighbors in space are considered outliers. Rather than utilizing factual or distance measures, deviation-based techniques distinguish exceptions/outlier by inspecting differences in the principle attributes of items in a group.

8. Genetic Algorithm

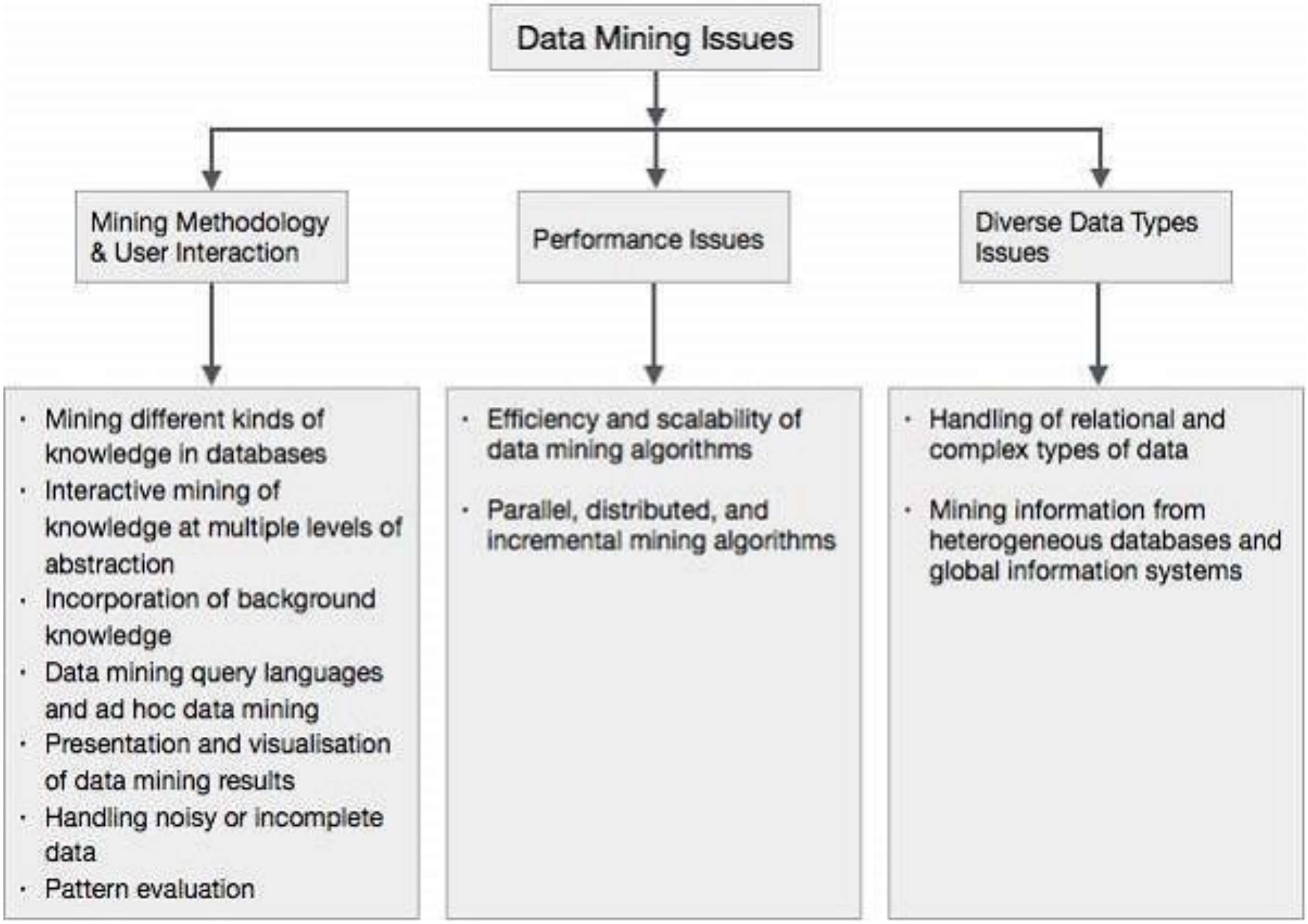
Genetic algorithms are adaptive heuristic search algorithms that belong to the larger part of evolutionary algorithms. Genetic algorithms are based on the ideas of natural selection and genetics. These are intelligent exploitation of random search provided with historical data to direct the search into the region of better performance in solution space. They are commonly used to generate high-quality solutions for optimization problems and search problems. Genetic algorithms simulate the process of natural selection which means those species who can adapt to changes in their environment are able to survive and reproduce and go to the next generation. In simple words, they simulate "survival of the fittest" among individuals of consecutive generations for solving a problem. Each generation consist of a population of individuals and each individual represents a point in search space and possible solution. Each individual is represented as a string of character/integer/float/bits. This string is analogous to the Chromosome.

Data mining Issues-

Data mining is not an easy task, as the algorithms used can get very complex and data is not always available at one place. It needs to be integrated from various heterogeneous data sources. These factors also create some issues. The major issues regarding −

- **Mining Methodology and User Interaction**
- **Performance Issues**
- **Diverse Data Types Issues**

The following diagram describes the major issues.



Mining Methodology and User Interaction Issues

It refers to the following kinds of issues −

- **Mining different kinds of knowledge in databases** − Different users may be interested in different kinds of knowledge. Therefore it is necessary for data mining to cover a broad range of knowledge discovery task.
- **Interactive mining of knowledge at multiple levels of abstraction** − The data mining process needs to be interactive because it allows users to focus the search for patterns, providing and refining data mining requests based on the returned results.
- **Incorporation of background knowledge** − to guide discovery process and to express the discovered patterns, the background knowledge can be used. Background knowledge may be used to express the discovered patterns not only in concise terms but at multiple levels of abstraction.
- **Data mining query languages and ad hoc data mining** − Data Mining Query language that allows the user to describe ad hoc mining tasks, should be integrated with a data warehouse query language and optimized for efficient and flexible data mining.

- **Presentation and visualization of data mining results** − Once the patterns are discovered it needs to be expressed in high level languages, and visual representations. These representations should be easily understandable.
- **Handling noisy or incomplete data** − the data cleaning methods are required to handle the noise and incomplete objects while mining the data regularities. If the data cleaning methods are not there then the accuracy of the discovered patterns will be poor.
- **Pattern evaluation** − the patterns discovered should be interesting because either they represent common knowledge or lack novelty.

Performance Issues

There can be performance-related issues such as follows −

- **Efficiency and scalability of data mining algorithms** − In order to effectively extract the information from huge amount of data in databases, data mining algorithm must be efficient and scalable.
- **Parallel, distributed, and incremental mining algorithms** − The factors such as huge size of databases, wide distribution of data, and complexity of data mining methods motivate the development of parallel and distributed data mining algorithms. These algorithms divide the data into partitions which is further processed in a parallel fashion. Then the results from the partitions is merged. The incremental algorithms, update databases without mining the data again from scratch.

Diverse Data Types Issues

- **Handling of relational and complex types of data** − the database may contain complex data objects, multimedia data objects, spatial data, temporal data etc. It is not possible for one system to mine all these kind of data.
- **Mining information from heterogeneous databases and global information systems** − the data is available at different data sources on LAN or WAN. These data source may be structured, semi structured or unstructured. Therefore mining the knowledge from them adds challenges to data mining.

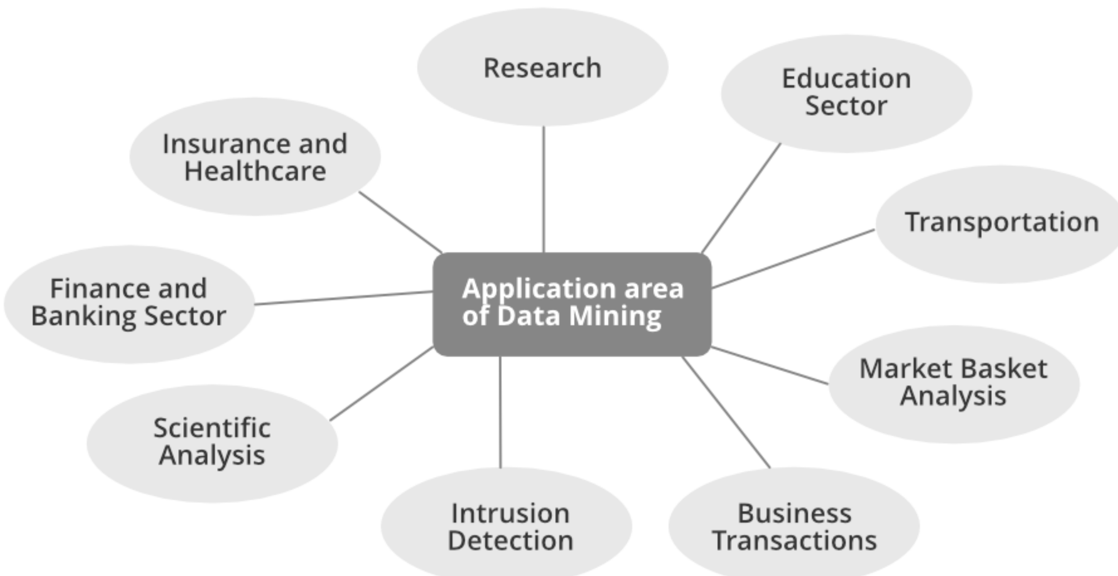
Applications of Data Mining

1. Financial Analysis
2. Biological Analysis
3. Scientific Analysis
4. Intrusion Detection
5. Fraud Detection
6. Research Analysis

Data mining is the computational process of analyzing data from different perspectives, dimensions, angles and categorizing/summarizing it into meaningful information. Data Mining can be applied to any type of data e.g. Data Warehouses, Transactional Databases, Relational Databases, Multimedia Databases, Spatial Databases, Time-series Databases, World Wide Web.

Data mining provides competitive advantages in the knowledge economy. It does this by providing the maximum knowledge needed to rapidly make valuable business decisions despite the enormous amounts of available data.

There are many measurable benefits that have been achieved in different application areas from data mining. So, let's discuss different applications of Data Mining:

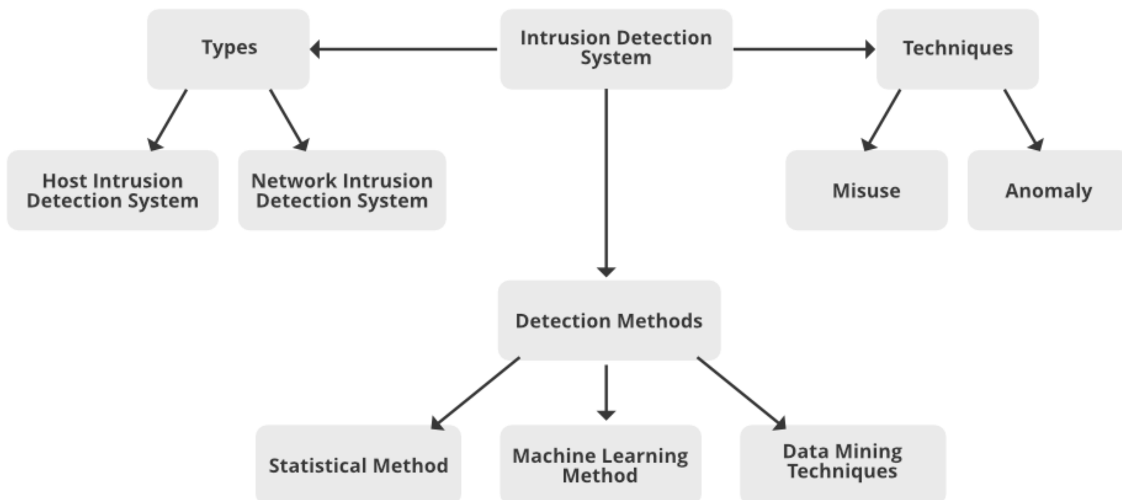


Scientific Analysis: Scientific simulations are generating bulks of data every day. This includes data collected from nuclear laboratories, data about human psychology, etc. Data mining techniques are capable of the analysis of these data. Now we can capture and store more new data faster than we can analyze the old data already accumulated. Example of scientific analysis:

- Sequence analysis in bioinformatics
- Classification of astronomical objects
- Medical decision support.

Intrusion Detection: A network intrusion refers to any unauthorized activity on a digital network. Network intrusions often involve stealing valuable network resources. Data mining technique plays a vital role in searching intrusion detection, network attacks, and anomalies. These techniques help in selecting and refining useful and relevant information from large data sets. Data mining technique helps in classify relevant data for Intrusion Detection System. Intrusion Detection system generates alarms for the network traffic about the foreign invasions in the system. For example:

- Detect security violations
- Misuse Detection
- Anomaly Detection



Business Transactions: Every business industry is memorized for perpetuity. Such transactions are usually time-related and can be inter-business deals or intra-business operations. The effective and in-time use of the data in a reasonable time frame for competitive decision-making is definitely the most important problem to solve for businesses that struggle to survive in a highly competitive world. Data mining helps to analyze these business transactions and identify marketing approaches and decision-making. Example :

- Direct mail targeting
- Stock trading
- Customer segmentation
- Churn prediction (Churn prediction is one of the most popular Big Data use cases in business)

Market Basket Analysis: Market Basket Analysis is a technique that gives the careful study of purchases done by a customer in a supermarket. This concept identifies the pattern of frequent purchase items by customers. This analysis can help to promote deals, offers, sale by the companies and data mining techniques helps to achieve this analysis task. Example:

- Data mining concepts are in use for Sales and marketing to provide better customer service, to improve cross-selling opportunities, to increase direct mail response rates.

- Customer Retention in the form of pattern identification and prediction of likely defections is possible by Data mining.
- Risk Assessment and Fraud area also use the data-mining concept for identifying inappropriate or unusual behavior etc.

Education: For analyzing the education sector, data mining uses Educational Data Mining (EDM) method. This method generates patterns that can be used both by learners and educators. By using data mining EDM we can perform some educational task:

- Predicting students admission in higher education
- Predicting students profiling
- Predicting student performance
- Teachers teaching performance
- Curriculum development
- Predicting student placement opportunities

Research: A data mining technique can perform predictions, classification, clustering, associations, and grouping of data with perfection in the research area. Rules generated by data mining are unique to find results. In most of the technical research in data mining, we create a training model and testing model. The training/testing model is a strategy to measure the precision of the proposed model. It is called Train/Test because we split the data set into two sets: a training data set and a testing data set. A training data set used to design the training model whereas testing data set is used in the testing model.

Example:

- Classification of uncertain data.
- Information-based clustering.
- Decision support system
- Web Mining
- Domain-driven data mining
- IoT (Internet of Things)and Cybersecurity
- Smart farming IoT(Internet of Things)

Healthcare and Insurance: A Pharmaceutical sector can examine its new deals force activity and their outcomes to improve the focusing of high-value physicians and figure out which promoting activities will have the best effect in the following upcoming months, Whereas the Insurance sector, data mining can help to predict which customers will buy new policies, identify behavior patterns of risky customers and identify fraudulent behavior of customers.

- Claims analysis i.e which medical procedures are claimed together.
- Identify successful medical therapies for different illnesses.

- Characterizes patient behavior to predict office visits.

Transportation: A diversified transportation company with a large direct sales force can apply data mining to identify the best prospects for its services. A large consumer merchandise organization can apply information mining to improve its business cycle to retailers.

- Determine the distribution schedules among outlets.
- Analyze loading patterns.

Financial/Banking Sector: A credit card company can leverage its vast warehouse of customer transaction data to identify customers most likely to be interested in a new credit product.

- Credit card fraud detection.
- Identify ‘Loyal’ customers.
- Extraction of information related to customers.
- Determine credit card spending by customer groups.

Data Objects and attribute types-

Data objects can also be referred to as samples, examples, instances, data points, or objects. If the data objects are stored in a database, they are data tuples. That is, the rows of a database correspond to the data objects, and the columns correspond to the attributes.

Data sets are made up of data objects. A **data object** represents an entity—in a sales database, the objects may be customers, store items, and sales; in a medical database, the objects may be patients; in a university database, the objects may be students, professors, and courses. Data objects are typically described by attributes. Data objects can also be referred to as *samples*, *examples*, *instances*, *data points*, or *objects*. If the data objects are stored in a database, they are *data tuples*. That is, the rows of a database correspond to the data objects, and the columns correspond to the attributes.

Data objects are the essential part of a database. A data object represents the entity. Data Objects are like a group of attributes of an entity. For example, a sales data object may represent customers, sales, or purchases. When a data object is listed in a database they are called data tuples.

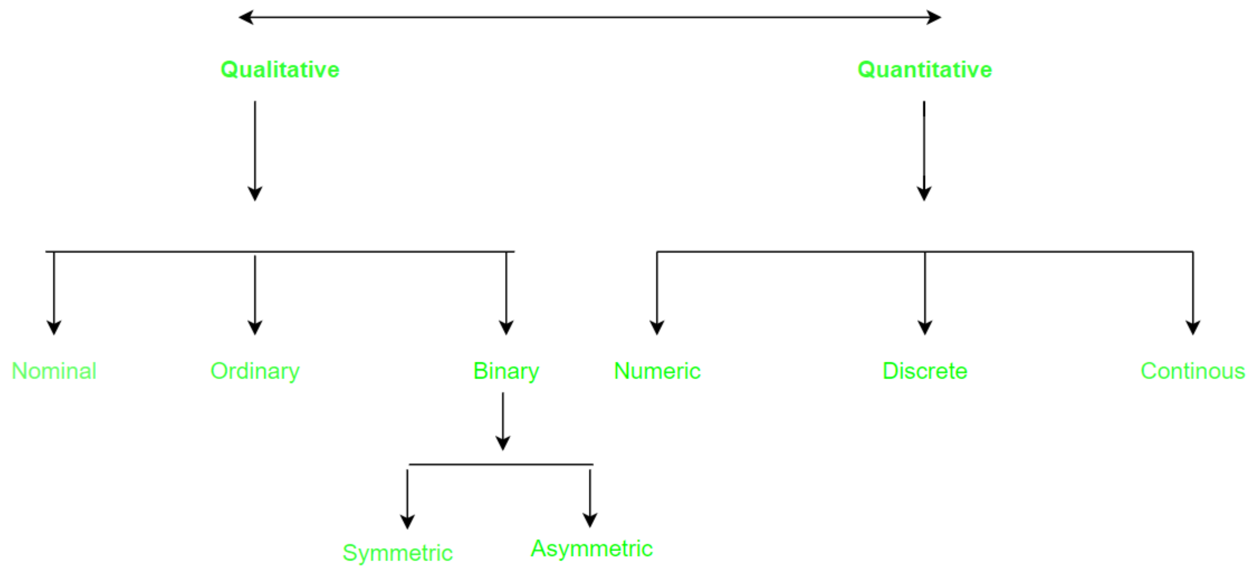
Attribute:

It can be seen as a data field that represents the characteristics or features of a data object. For a customer, object attributes can be customer Id, address, etc. We can say that a **set of attributes used to describe a given object are known as attribute vector or feature vector.**

Type of attributes :

This is the First step of Data-preprocessing. We differentiate between different types of attributes and then preprocess the data. So here is the description of attribute types.

1. Qualitative (Nominal (N), Ordinal (O), Binary(B)).
2. Quantitative (Numeric, Discrete, Continuous)



Qualitative Attributes:

1. Nominal Attributes – related to names: The values of a Nominal attribute are names of things, some kind of symbols. Values of Nominal attributes represents some category or state and that's why nominal attribute also referred as **categorical attributes** and there is no order (rank, position) among values of the nominal attribute.

Example :

Attribute	Values
Colours	Black, Brown, White
Categorical Data	Lecturer, Professor, Assistant Professor

2. Binary Attributes: Binary data has only 2 values/states. For Example yes or no, affected or unaffected, true or false.

- **Symmetric:** Both values are equally important (Gender).
- **Asymmetric:** Both values are not equally important (Result).

Attribute	Values
Gender	Male , Female

Attribute	Values
Cancer detected	Yes, No
result	Pass , Fail

3. Ordinal Attributes : The Ordinal Attributes contains values that have a meaningful sequence or ranking(order) between them, but the magnitude between values is not actually known, the order of values that shows what is important but don't indicate how important it is.

Attribute	Value
Grade	A,B,C,D,E,F
Basic pay scale	16,17,18

Quantitative Attributes:

1. Numeric: A numeric attribute is quantitative because, it is a measurable quantity, represented in integer or real values. Numerical attributes are of 2 types, **interval**, and **ratio**.

- An **interval-scaled** attribute has values, whose differences are interpretable, but the numerical attributes do not have the correct reference point, or we can call zero points. Data can be added and subtracted at an interval scale but can not be multiplied or divided. Consider an example of temperature in degrees Centigrade. If a day's temperature of one day is twice of the other day we cannot say that one day is twice as hot as another day.
- A **ratio-scaled** attribute is a numeric attribute with a fix zero-point. If a measurement is ratio-scaled, we can say of a value as being a multiple (or ratio) of another value. The values are ordered, and we can also compute the difference between values, and the mean, median, mode, Quantile-range, and Five number summary can be given.

2. Discrete : Discrete data have finite values it can be numerical and can also be in categorical form. These attributes has finite or countably infinite set of values.

Example:

Attribute	Value
Profession	Teacher, Business man, Peon
ZIP Code	301701, 110040

3. Continuous: Continuous data have an infinite no of states. Continuous data is of float type. There can be many values between 2 and 3.

Example :

Attribute	Value
Height	5.4, 6.2 ...etc
weight	50.33 ....etc

Statistical description of data-

Descriptive statistics are like a translator for data that helps provide the patterns and trends of data. Descriptive statistics use various tools, such as measures of central tendency and variability and graphical representations, to present data in a meaningful and accessible way. Central tendency measures the typical value of a dataset, while variability measures show how well the spread-out data is. Graphical representations add a visual element, making it easier to spot patterns and trends. Descriptive statistics are essential for analyzing and understanding data, providing a foundation for complex statistical analysis in various fields, including business, healthcare, and social sciences.

Bar graphs, pie charts, and line graphs may be found in most statistical or graphical data presentation software programs. Quantile plots, quantile-quantile plots, histograms, and scatter plots are more common methods for displaying data summaries and distributions.

Types of Descriptive Statistics

Descriptive Statistics can be divided in terms of data types, patterns, or characteristics- Distribution(or frequency distribution), Centra Tendency, and Variability(or Dispersion)

Frequency Distribution

The number of times a specific value appears in the data is its frequency (f). A variable's distribution is its frequency pattern or the collection of all conceivable values and the frequencies corresponding to those values. Frequency tables or charts are used to represent frequency distributions. Both categorical and numerical variables can be employed using frequency distribution tables. Only class intervals, which will be detailed momentarily, should be used with continuous variables. Bar graphs, pie charts, and line graphs may be found in most statistical or graphical data presentation software programs. Quantile plots, quantile-quantile plots, histograms, and scatter plots are more common methods for displaying data summaries and distributions.

Central Tendency Indicators: Mean, Median, and Mode

Mean

It describes how to calculate the average of a set of data using various methods. Let's imagine that we have a database that stores object with values for attributes such as salary, and let's also pretend that we have access to this database.

Consequently, the N observations that we have of X are represented as follows: x1, x2,... xN. Within the scope of this discussion, the compilation of numerical information is typically referred to as the "data set" (for X). On what part of the scatter plot would the majority of the salary data points be located? Because of this, we are able to gain a sense of the general pattern of the data.

Indicators of central tendency include the mean, median, mode, and even the data's midpoint.

The (arithmetic) mean is the most common and trustworthy numerical measure of the "center" of a data set. It is also one of the most often used measures of central tendency. Let's assume that X is a numeric property and that x1, x2,..., and xN are the N values or observations. The sum total of these numbers is equal to

MEAN $= \sum N \div n$

where N is the sum of all integers

n represents the total number of integers.

DATA: { 16, 17, 10, 13, 20, 18, 13, 14, 18 }

$$\bar{x} = \frac{16 + 17 + 10 + 13 + 20 + 18 + 13 + 14 + 18}{9}$$

$$\bar{x} = \frac{139}{9}$$

$\bar{x} = 15.444$ (rounded to three decimal places)

Median

While there is no more descriptive statistic than the mean, it is not necessarily the most accurate approach to locating the data center. The mean's extreme value (or outlier) sensitivity is a serious flaw. Even a handful of outliers can skew the median. For instance, some companies may have one or two exceptionally well-paid managers who drive up the average income for everyone else. The same holds true for test results: a small number of students with extremely poor marks can significantly lower the class average. The trimmed mean, which is the mean found by cutting off the values at the high and low ends, can be used to reduce the effect of a tiny number of extreme values. In order to find the average income, for instance, we may filter the data and discard the highest and lowest numbers. Avoiding a 20% snip at each end might save us from losing some crucial details.

The median, the middle value in a set of ordered data values, is a more appropriate measure of the center of data when the data are skewed (asymmetric). It's the threshold that divides the top half from the bottom half of a dataset.

While the median is typically used with numerical data in probability and statistics, applying the notion to ordinal data is also possible. For the sake of argument, let's say that N values for attribute X have been sorted ascendingly in a given data collection. The median of an ordered collection is the middle value if N is odd. In the case where N is an even number, the median is not a discrete value but rather the midpoint between the two extremes.

The median is calculated as the mean of the middle two values of a numerical attribute X.

Mode

The mode, in addition to the mean, can be utilized in calculating the average. The mode is the statistic representing the value that occurs the most frequently in a set of numbers.

As a consequence of this, it is possible to compute both its qualitative and quantitative features of it. It is possible for there to be a variety of modes since the frequency with the highest value might have a number of different values. Data sets can be classified as unimodal, bimodal, or trimodal, depending on the number of modes they include. When talking about multimodal data sets, it is usual practice to refer to them as having two or more modes. On the other hand, if each data value only appears once, there will be no mode.

$$\text{Mean} = \frac{\text{Sum of Observations}}{\text{Total Number of Observations}}$$

If 'n' is odd: $$\text{Median} = \left(\frac{n+1}{2}\right)^{th} \text{term}$$

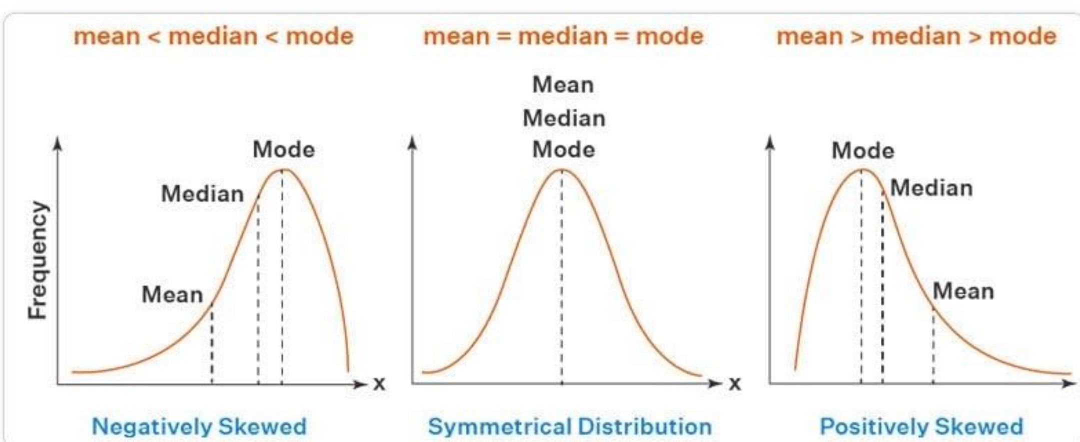
If 'n' is even: $$\text{Median} = \frac{\left(\frac{n}{2}\right)^{th}\text{term} + \left(\frac{n}{2}+1\right)^{th}\text{term}}{2}$$

$$\text{Mode} = L + h\,\frac{(f_m - f_1)}{(f_m - f_1) + (f_m - f_2)}$$

To know why and how to pursue a career in data science, refer to the data science career path.

The following empirical connection holds for unimodal frequency distributions that are somewhat asymmetrical:

Mean−Mode = 3×(Mean−Median)



As a result, when the mean and median are available, calculating the mode of a unimodal frequency distribution that is only slightly skewed is a piece of cake.

As shown in the Figure above, the mean, median, and mode are all centered on the same value in a unimodal frequency curve exhibiting perfect symmetry in the data. Unfortunately, though, the facts in most practical contexts are asymmetric. It's also possible for them to be positively skewed, with the mode below the median, or negatively skewed, with the mode above the median.

The central tendency of data collection may also be evaluated using the midrange. It is calculated by averaging the greatest and lowest numbers in the set. SQL's max() and min() aggregate methods make it simple to calculate this algebraic metric ().

A lot of people are confused about the role of a Data Scientist and a Data Analyst; even though both deal with "*Data*," there are still a good number of significant differences between them. Do you want to know the clear difference between a data scientist and a data analyst, click here.

The Dispersion of Data

Before diving into further measures of data dispersion, let's first examine the range, quantiles, quartiles, percentiles, and interquartile range.

Range

Let X be a numerical attribute, and let x1, x2,..., xN be a series of observations for X. The set's range is calculated by taking the absolute value of the difference between the greatest and smallest values in the set (max()-min()).

$$\textbf{Range} = X_{max} - X_{min}$$

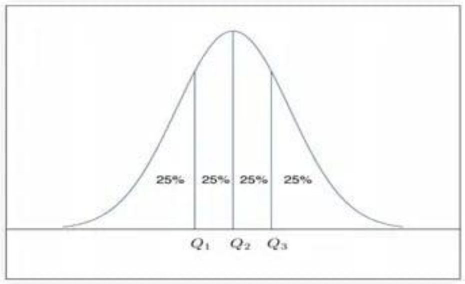
Quartile

Let's pretend that we have a set of numbers ordered from largest to smallest for attribute X. Imagine we could select particular data points to partition the data distribution equally. Quartiles are a statistical description of data measures based on these values. The quartiles of distribution are discrete numbers chosen at predetermined intervals to create groups of data that are roughly equivalent in size. (We say "*basically*" because there might not be values of X that divide the data into precisely equal-sized sections. For the sake of clarity, let's assume they're on par. For a given distribution of data, the kth q-quartile is the number x such that no more than k/q of the data values are fewer than x, and no more than (q k)/q of the data values are higher than x, where k is an integer such that 0 k q. Each of the q-quartiles has a probability of q-1.

The 2-quartile is the midpoint between a data sets minimum and maximum values. You may think of it as the middle number. The three data points known as the "*4-quartiles*" that divide the distribution into four equal parts, with each portion representing one-fourth of the whole, provide the basis for this analysis. Most people will refer to them as "*quartiles*" instead. The median, the quartiles, and the percentiles are the most often used quartiles, whereas the 100-quartiles are more usually known as percentiles; they divide the data distribution into 100 equal-sized sequential groupings.

The quartiles can be used to determine a distribution's shape, size, and central tendency. Q1 represents the 25th percentile or the first quartile. It discards the bottom 25% of the records.

In this case, Q3 represents the 75th percentile, which eliminates the bottom 75% (or top 25%) of the data. The 50th percentile is the middle quartile. The median represents the midpoint of a set of data.



A straightforward measure of dispersion, the gap between the first and third quartiles reveals the span of values occupied by the middle 50% of the data. The IQR formula is IQR = Q3 Q1, where Q3 and Q1 are the third and first quartiles.

$$\text{Quartile Deviation} = \frac{\text{Third Quartile} - \text{First Quartile}}{2}$$

$$\text{Quartile Deviation } (Q.D) = \frac{Q_3 - Q_1}{2}$$

Variance and Standard Deviation

Distributive metrics of data include variance and standard deviation. They show how dispersed a data set is. When the standard deviation is small, the data points cluster tightly around the mean, but when it's big, the data points are dispersed over a wide range of values.

The variance of N observations, x1,x2,...,xN, for a numeric attribute X

The **variance** of a population of n is $\sigma^2 = \frac{\sum_{i=1}^{n}(x_i - \mu)^2}{n} = \frac{\sum_{i=1}^{n} x_i^2}{n} - \mu^2.$

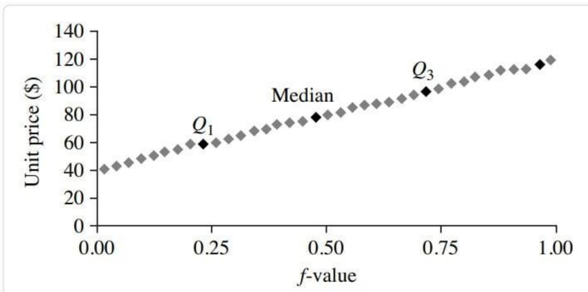
The **standard deviation** of a population of n is $\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{n}}.$

A Quantile-Quantile Plot

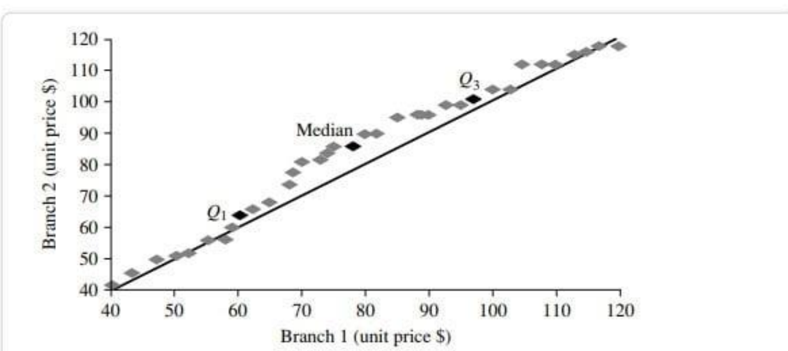
The quantiles of one uni variate distribution are shown against the quantiles of another distribution in a q-q graphic. It's an effective mental imagery method since it permits the user to see whether there is a change while switching distributions.

Let's pretend we have data from two branches with respect to the attribute or variable unit price. For simplicity, we will refer to the first branch's data as x1,...,xN and the second branch's data as y1,...,yM, with both sets of data being arranged from most recent to least recent. In the case where M = N (i.e., the number of points in both sets is equal), we may plot yi versus xi, where yi and xi are the I 0.5)/N quantiles of the two sets of data.

Only M points will fit on the q-q plot if M N (the second branch contains fewer data than the first). The quantile of the y distribution at index I, expressed as yi, is given by I 0.5)/M.



Unit price ($)	Count of items sold
40	275
43	300
47	250
—	—
74	360
75	515
78	540
—	—
115	320
117	270
120	350



Distribution of the quantiles vs the quantiles. Plots unit pricing data on a quantile-quantile retail sales chart at two All Electronics locations during a specific time frame. For each data set, each dot represents a quantile, and the unit prices at branch 1 are shown against those at branch 2 at that quantile. (The straight line is meant to be compared to when the unit price is the same at all branches for a particular quantile. The darker data points represent information from Quarter 1, the median, and Quarter 3.For example, we can observe that in the first quarter of this year, the unit price of things sold at branch 1 was somewhat lower than that at branch 2. So, 25% of sales at Store 1 were for products that cost less than equivalent to $60, but at Branch 2, only 25% of things sold were $60 or less. Half of the things sold at Branch 1 were less than $78 (the median, see Q2), and half sold at Branch 2 were less than $85 (the 25th percentile). Generally, we see that the unit prices of goods sold at branch 1 are lower than those sold at branch 2, indicating a change in distribution.

Histograms

The histogram, sometimes known as a frequency histogram, has been used for at least a century. The word "*histogram*" comes from the Greek words for "*pole*" and "*gramme*," therefore, a histogram is literally a chart of poles. Histograms are a graphical representation for summing up the distribution of an individual value of an attribute X. For each known value, a pole or vertical bar is drawn if X is nominal, like a car model or item category. Each bar's height represents the number of times that particular X value occurred. This type of chart is most generally referred to as a bar chart.

Histogram is the recommended phrase if X is a numerical variable. Separate, sequential subranges of X values have been established. The subranges represent separate segments of X's data distribution, sometimes known as buckets or bins. A bucket's breadth refers to its reach. In most cases, each bucket has the same width as the others. A price characteristic with values from $1 (rounded up to the closest dollar) to $200 (also rounded up to the nearest dollar) can be divided into the subranges 1-20, 21-40, 42-60, etc.

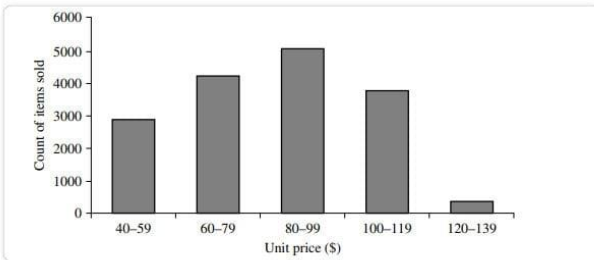
Each subrange is represented by a bar whose height is proportional to the number of data points that fall within that subrange.

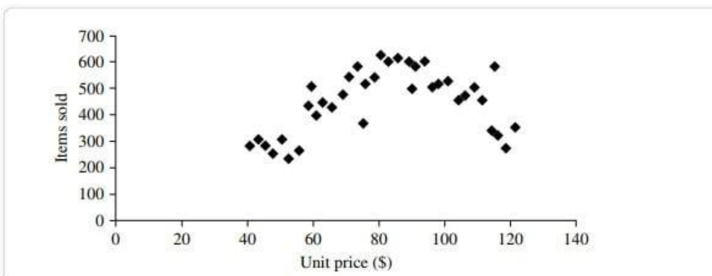
Data Correlation and Scatter Plots

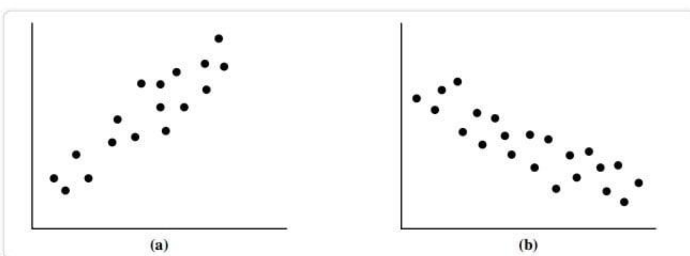
In order to determine if there is a correlation between two variables, a scatter plot is a useful graphical tool. Apparent connection, pattern, or development between the two quantitative features. Each pair of values is interpreted as an algebraic pair of coordinates and represented as a point on the graph paper. The data in Figure are plotted as a scatter diagram. Table

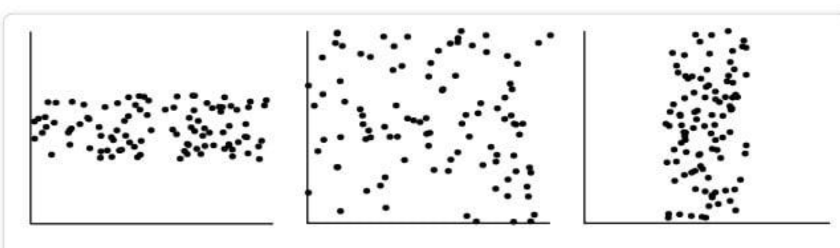
The scatter plot is an effective tool for quickly seeing patterns in bivariate data, identifying outliers, and investigating possible correlations.

When one characteristic (X) implies the other (Y), we say the two characteristics are correlated. A correlation can be either positive or negative or even nonexistent (uncorrelated). Examples of positive and negative relationships between the two characteristics are displayed in the figure. When a pattern of points is plotted, a slope is indicated if the points. Since X is increasing while Y is increasing, a positive correlation may be inferred from the data (Figure). Given a plotted pattern,









Slopes from left to right, indicating a negative association between X and Y; X values rise as Y values fall (Figure). A line of best fit can be drawn to investigate the link between the variables. You may find information about correlation tests using statistics. In the data sets depicted in each figure, there are three instances in which a correlation exists between the two qualities.

Data Preprocessing – Cleaning, Integration, Reduction, Transformation and discretization, Data Visualization,

Data preprocessing is an important step in the data mining process. It refers to the cleaning, transforming, and integrating of data in order to make it ready for analysis. The goal of data preprocessing is to improve the quality of the data and to make it more suitable for the specific data mining task.

Some common steps in data preprocessing include:

Data preprocessing is an important step in the data mining process that involves cleaning and transforming raw data to make it suitable for analysis. Some common steps in data preprocessing include:

Data Cleaning: This involves identifying and correcting errors or inconsistencies in the data, such as missing values, outliers, and duplicates. Various techniques can be used for data cleaning, such as imputation, removal, and transformation.

Data Integration: This involves combining data from multiple sources to create a unified dataset. Data integration can be challenging as it requires handling data with different formats, structures, and semantics. Techniques such as record linkage and data fusion can be used for data integration.

Data Transformation: This involves converting the data into a suitable format for analysis. Common techniques used in data transformation include normalization, standardization, and discretization. Normalization is used to scale the data to a common range, while standardization is used to transform the data to have zero mean and unit variance. Discretization is used to convert continuous data into discrete categories.

Data Reduction: This involves reducing the size of the dataset while preserving the important information. Data reduction can be achieved through techniques such as feature selection and feature extraction. Feature selection involves selecting a subset of relevant features from the dataset, while feature extraction involves transforming the data into a lower-dimensional space while preserving the important information.

Data Discretization: This involves dividing continuous data into discrete categories or intervals. Discretization is often used in data mining and machine learning algorithms that require categorical data. Discretization can be achieved through techniques such as equal width binning, equal frequency binning, and clustering.

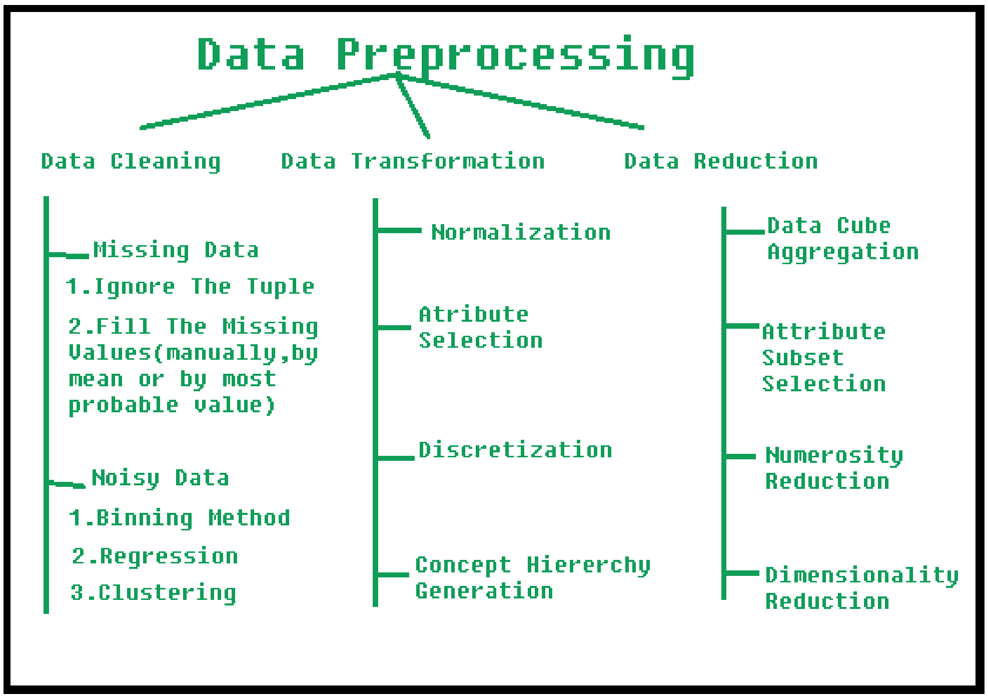
Data Normalization: This involves scaling the data to a common range, such as between 0 and 1 or -1 and 1. Normalization is often used to handle data with different units and scales. Common normalization techniques include min-max normalization, z-score normalization, and decimal scaling.

Data preprocessing plays a crucial role in ensuring the quality of data and the accuracy of the analysis results. The specific steps involved in data preprocessing may vary depending on the nature of the data and the analysis goals.

By performing these steps, the data mining process becomes more efficient and the results become more accurate.

Preprocessing in Data Mining:

Data preprocessing is a data mining technique which is used to transform the raw data in a useful and efficient format.



Steps Involved in Data Preprocessing:

1. Data Cleaning:

The data can have many irrelevant and missing parts. To handle this part, data cleaning is done. It involves handling of missing data, noisy data etc.

(a). Missing Data:

This situation arises when some data is missing in the data. It can be handled in various ways. Some of them are:

1) **Ignore the tuples:** This approach is suitable only when the dataset we have is quite large and multiple values are missing within a tuple.
2) **Fill the Missing values:** There are various ways to do this task. You can choose to fill the missing values manually, by attribute mean or the most probable value.

(b). Noisy Data:

Noisy data is a meaningless data that can't be interpreted by machines. It can be generated due to faulty data collection, data entry errors etc. It can be handled in following ways :

1) **Binning Method:** This method works on sorted data in order to smooth it. The whole data is divided into segments of equal size and then various methods are performed to complete the task. Each segmented is handled separately. One can replace all data in a segment by its mean or boundary values can be used to complete the task.
2) **Regression:** Here data can be made smooth by fitting it to a regression function. The regression used may be linear (having one independent variable) or multiple (having multiple independent variables).
3) **Clustering:** This approach groups the similar data in a cluster. The outliers may be undetected or it will fall outside the clusters.

2. Data Transformation: This step is taken in order to transform the data in appropriate forms suitable for mining process. This involves following ways:

1) **Normalization:** It is done in order to scale the data values in a specified range (-1.0 to 1.0 or 0.0 to 1.0)
2) **Attribute Selection:** In this strategy, new attributes are constructed from the given set of attributes to help the mining process.
3) **Discretization:** This is done to replace the raw values of numeric attribute by interval levels or conceptual levels.
4) **Concept Hierarchy Generation:** Here attributes are converted from lower level to higher level in hierarchy. For Example-The attribute "city" can be converted to "country".

3. Data Reduction: Data reduction is a crucial step in the data mining process that involves reducing the size of the dataset while preserving the important information. This is done to improve the efficiency of data analysis and to avoid overfitting of the model. Some common steps involved in data reduction are:

Feature Selection: This involves selecting a subset of relevant features from the dataset. Feature selection is often performed to remove irrelevant or redundant features from the dataset. It can be done using various techniques such as correlation analysis, mutual information, and principal component analysis (PCA).

Feature Extraction: This involves transforming the data into a lower-dimensional space while preserving the important information. Feature extraction is often used when the original features are high-dimensional and complex. It can be done using techniques such as PCA, linear discriminant analysis (LDA), and non-negative matrix factorization (NMF).

Sampling: This involves selecting a subset of data points from the dataset. Sampling is often used to reduce the size of the dataset while preserving the important information. It can be done using techniques such as random sampling, stratified sampling, and systematic sampling.

Clustering: This involves grouping similar data points together into clusters. Clustering is often used to reduce the size of the dataset by replacing similar data points with a representative centroid. It can be done using techniques such as k-means, hierarchical clustering, and density-based clustering.

Compression: This involves compressing the dataset while preserving the important information. Compression is often used to reduce the size of the dataset for storage and transmission purposes. It can be done using techniques such as wavelet compression, JPEG compression, and gzip compression.

Data similarity and dissimilarity measures:

Measuring similarity and dissimilarity in data mining is an important task that helps identify patterns and relationships in large datasets. To quantify the degree of similarity or dissimilarity between two data points or objects, mathematical functions called similarity and dissimilarity measures are used. Similarity measures produce a score that indicates the degree of similarity between two data points, while dissimilarity measures produce a score that indicates the degree of dissimilarity between two data points. These measures are crucial for many data mining tasks, such as identifying duplicate records, clustering, classification, and anomaly detection.

Basics of Similarity and Dissimilarity Measures

Similarity Measure

- A similarity measure is a mathematical function that quantifies the degree of similarity between two objects or data points. It is a numerical score measuring how alike two data points are.
- It takes two data points as input and produces a similarity score as output, typically ranging from 0 (completely dissimilar) to 1 (identical or perfectly similar).
- A similarity measure can be based on various mathematical techniques such as Cosine similarity, Jaccard similarity, and Pearson correlation coefficient.
- Similarity measures are generally used to identify duplicate records, equivalent instances, or identifying clusters.

Dissimilarity Measure

- A dissimilarity measure is a mathematical function that quantifies the degree of dissimilarity between two objects or data points. It is a numerical score measuring how different two data points are.
- It takes two data points as input and produces a dissimilarity score as output, ranging from 0 (identical or perfectly similar) to 1 (completely dissimilar). A few dissimilarity measures also have infinity as their upper limit.

- A dissimilarity measure can be obtained by using different techniques such as Euclidean distance, Manhattan distance, and Hamming distance.
- Dissimilarity measures are often used in identifying outliers, anomalies, or clusters.

Data Types Similarity and Dissimilarity Measures

- For nominal variables, these measures are binary, indicating whether two values are equal or not.
- For ordinal variables, it is the difference between two values that are normalized by the max distance. For the other variables, it is just a distance function.

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